Softmax Numerics and Attention Sinks in LLM Serving: A Reproduction and Measurement Study

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Abstract

Serving engines compute softmax in every attention head and in the vocabulary head. We re-test four published claims about it: dispersion with growing context and an adaptive-temperature fix, first-token attention sinks, Softmax Collapse, and Softmax Cancellation. We use fine-grid CPU reproductions and the papers' own published code. The published collapse cliffs (5.5, 16.6, 36.7 nats) are two-class values: with $\geq$113 classes, CPU PyTorch collapses up to 2.08 nats earlier. The published adaptive-temperature polynomial acts only on rows with entropy between 0.849 and 5.945 nats. In a matched 340M softmax/softpick pair, the softmax model's sink rate falls from 62.6% to 10.6% between 256 and 4,096 tokens, while softpick shows none. We also locate where vLLM and SGLang compute returned log-probabilities; SGLang's RL on-policy mode uses bfloat16. Code, data and validation logs are released.

Working draft, November 4, 2026. Real-model results cover the models whose runs have finished; the others, E7 on real text and the GPU experiments (B4–B6) are marked [**Pending:** ...]. Set `\draftfalse` to remove line numbers and this box.

1 Introduction

A served language model evaluates softmax in every attention head and again in the vocabulary head. Four recent claims about it matter for serving:

- global attention must disperse as context grows, and an entropy-based adaptive temperature partly counters this (Veličković et al., 2025);
- the first-token attention sink is a learned mechanism that limits over-mixing (Barbero et al., 2025);
- standard cross-entropy has a precision cliff beyond which the target gradient is exactly zero, *Softmax Collapse* (Prieto et al., 2025; Barron et al., 2026);
- soft-target cross-entropy leaves a non-zero gradient when student and teacher agree, *Softmax Cancellation* (Barron et al., 2026).

Serving engines inherit these behaviours. Eviction policies decide which keys to keep (Xiao et al., 2024), kernels fold learned sinks into the normaliser (Dao-AILab, 2026), and RL trainers consume returned log-probabilities (SGLang project, 2026).

This paper re-tests the four claims and measures the related behaviour in five open models. It proposes no new method. Following the reproducibility-report format of TMLR and the ML Reproducibility Challenge (Fletcher et al., 2024), each claim is stated with its source, tested, and given a verdict. Following measurement papers on LLM serving (Agrawal et al., 2024; Chandrasekar & Kramberger, 2026), we report where a standard computation returns a misleading number. Our contributions:

1. a reproduction of four claims on CPU PyTorch, with fine grids, float64 and 50-digit references, and the papers' published code translated line by line (§4);

Table 1: Claims, sources and verdicts (details in §4–5).

#	Claim (paraphrased)	Source	Verdict
C1a	With bounded logits, every global-attention coefficient is $\Theta(1/n)$	Veličković et al. (2025, Thm. 2.2)	Fixed heads disperse; other heads take over (3 of 5 models run)
C1b	Entropy-based adaptive temperature sharpens attention at inference	Veličković et al. (2025, Fig. 4)	Holds inside the fitted regime; inactive above 5.945 nats
C2	First-token sinks are common; non-normalised attention removes them	Barbero et al. (2025); Gu et al. (2025); Zuhri et al. (2025)	Holds in the matched pair
C3	Target gradient is zero beyond $c \approx 5.5$ (bf16), 16.6 (fp32), 36.7 (fp64)	Barron et al. (2026, §1–2)	Holds for C=2; earlier for $C \geq 113$
C4	Non-zero gradient at z_S=z_T; the fix returns 0	Barron et al. (2026, §3)	Holds

2. real-model measurements of sink strength and planted-needle retrieval in five open models, including a matched softmax/softpick pair (Zayd M. K. Zuhri, 2026), with every probe validated bit for bit against the reference attention (§5);
3. an audit of how vLLM and SGLang compute returned log-probabilities (§6);
4. scripts for the remaining GPU experiments, one of them with a pre-registered success criterion (§9).

2 Claims checked

3 Methodology

Synthetic reproductions. All of these ran on CPU PyTorch 2.15.0, so every threshold below is a property of CPU kernels.

- **C3:** C classes, target logit at gap g above all others, built-in `F.cross_entropy`. We scan g on a 0.01 grid refined to 0.001 near each transition, and report the paper's margin $m = z_t - \mathrm{LSE}_{j \neq t} z_j = g - \ln(C-1)$.
- **C4:** 1,000 classes. The reference is float64 $q - p$ on the same rounded logits, checked against mpmath at 50 digits.
- **The fixes** are line-by-line translations of the paper's JAX listings.

Real-model probes. Five models run in float32 on CPU (Table 2). We register an attention function with the transformers attention interface. It computes exact softmax (or softpick) attention in query chunks and records statistics as it runs.

- **Validation:** logits match transformers' eager attention bit for bit for all five models. For the matched pair they also match an independent forward pass written from the authors' code.
- **Sensitivity:** swapping softpick for softmax in that reference changes the logits by up to 30.9, so the check does detect errors.
- **Loading:** perplexity on 1,024 held-out tokens is 17–40 for all five models.

Metrics. Probe statistics use the last 64 query positions of each sequence.

- **Sink rate:** the share of heads whose mean weight on token 0 exceeds 0.3 (Gu et al., 2025).

Table 2: Models probed. Perplexity on 1,024 tokens of Project Gutenberg #84, eager attention.

Model	Layers × heads	Attention	Context probed	PPL
Qwen3-0.6B	28 × 16 (8 KV)	global, QK-norm	256–8,192	31.6
Gemma-3-1B-it	26 × 4 (1 KV)	5 local (512) : 1 global	256–8,192	39.9
SmolLM2-360M	32 × 15 (5 KV)	global	256–8,192	17.0
vanilla-340M	24 × 16	global, softmax	256–4,096	17.3
softpick-340M	24 × 16	global, softpick	256–4,096	17.1

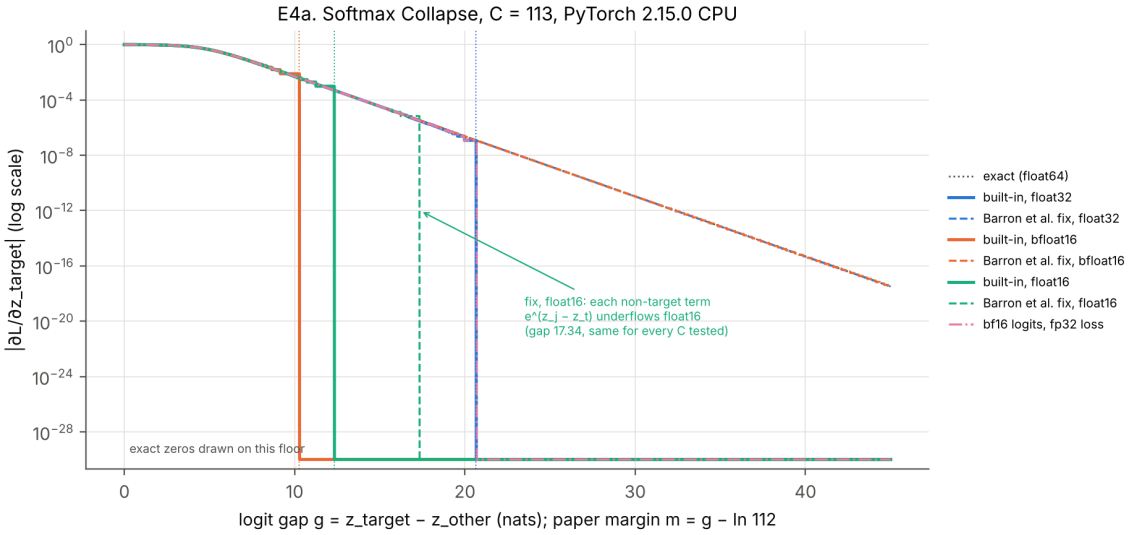


Figure 1: Target-logit gradient vs. logit gap, C=113, CPU PyTorch 2.15.0. Solid lines are built-in; dashed lines are the translated fix; dash-dot is bf16 logits with an fp32 loss. Exact zeros are drawn on the floor.

- **Needle weight:** the final query's weight on a planted sentence's answer tokens. We report it for the best head at each N, and for a fixed head (the best head at N=256). Under the fixed-gap law $w = e^{\Delta}/(e^{\Delta} + N - 1)$, the fitted gap $\Delta_N = \text{logit}(w) + \ln(N-1)$ stays constant.
- **Sampling:** three Gutenberg passages and three needle depths per point.

4 Reproduction results

4.1 C3: Softmax Collapse

Table 3: Precision cliff in the paper's margin units: published value vs. first exact zero on CPU PyTorch.

dtype	published	C=2	C=113	C=1000	C=151,936
bfloat16	≈5.5	5.547	5.564	5.562	5.507
float16	–	7.624	7.622	7.621	7.593
float32	≈16.6	16.636	15.943	15.944	15.943
float64	≈36.7	36.737	34.658	34.659	34.658

Verdict: holds, with a class-count condition (Table 3, Figure 1).

- The published cliffs are the two-class values $\ln(2/\epsilon)$.
- With ≥ 113 classes, CPU PyTorch collapses earlier: by 0.69 nats in fp32 and 2.08 nats in fp64. bf16 and fp16 stay within 0.04 of $\ln(2/\epsilon)$.

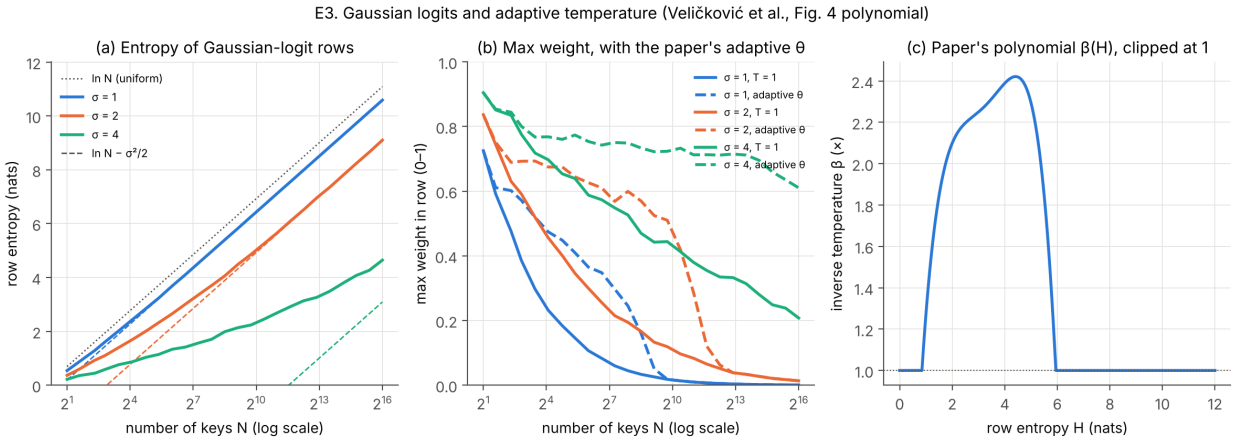


Figure 2: Gaussian-logit rows: entropy, maximum weight with and without the paper's adaptive θ, and the polynomial $\beta(H)$ clipped at 1.

Table 4: Sinks and planted-needle retrieval (last 64 queries; 3 passages and 3 needle depths per length).

model	$N_{\max}$	sink rate (256)	sink rate ($N_{\max}$)	fixed-head w (256→ $N_{\max}$)	best-head w ($N_{\max}$)	top-1
Qwen3-0.6B	*pending*					
Gemma-3-1B-it	*pending*					
SmolLM2-360M	8,192	73.3%	41.0%	0.961→0.0923	0.916	18/18
vanilla-340M	4,096	62.6%	10.6%	0.782→0.0019	0.767	15/15
softpick-340M	4,096	0.0%	0.0%	0.690→0.0737	0.635	14/15

- The fix never returns a zero target gradient in fp32 or bf16 (maximum relative error 3.2×10^{-7} and 9.7×10^{-3}). In fp16 it returns zero from gap 17.336 for every C tested, because each $e^{z_j - z_t}$ underflows there.
- With bf16 logits and an fp32 loss, the bf16 cliff disappears and the fp32 cliff remains.

4.2 C4: Softmax Cancellation

Verdict: holds. At $z_S = z_T$ the built-in gradient norm is 9.14×10^{-9} (fp32), 1.22×10^{-3} (bf16) and 2.27×10^{-4} (fp16); the fix returns exactly 0. Two storage choices drive the bf16 floor: the student's log-softmax stored in bf16 contributes 1.08×10^{-3}, and the teacher's probabilities stored in bf16 contribute 3.29×10^{-4}. With fp32 teacher probabilities and an upcast student, the floor is 2.2×10^{-8}.

4.3 C1b: Adaptive temperature

Verdict: holds inside the fitted regime; inactive beyond it (Figure 2).

- The printed polynomial gives $\beta > 1$ only for $H \in [0.849, 5.945]$ nats, peaking at $\beta = 2.42$.
- A target planted 4σ above Gaussian distractors rises from 0.125 to 0.798 of the row at $N = 235$.
- The correction switches off from $N = 861$. A uniform row over 4,096 keys already has 8.3 nats, so the fit needs refitting before use on long-context heads, as the authors' own description of it as ad hoc suggests (Veličković et al., 2025).

5 Real-model measurements

C2: sinks. In vanilla-340M, 62.6% of heads are sinks at 256 tokens, close to the 63.41% reported for the authors' 340M softmax model (Zuhri et al., 2025). The rate falls to 10.6% at 4,096 tokens. Gu et al. also

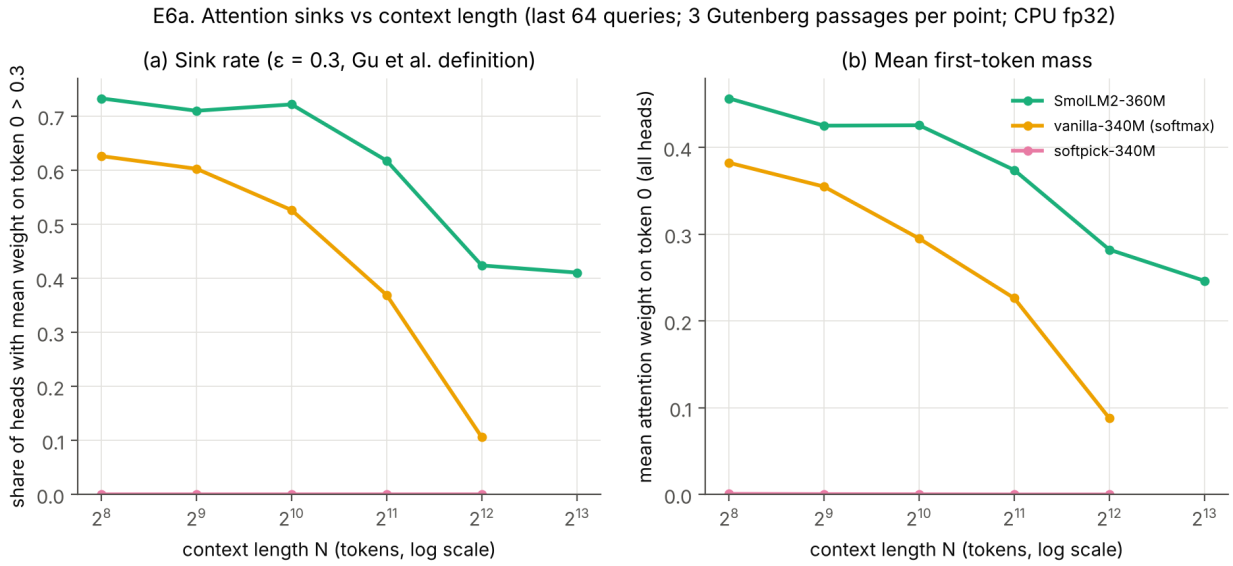


Figure 3: Sink rate and mean first-token mass vs. context length.

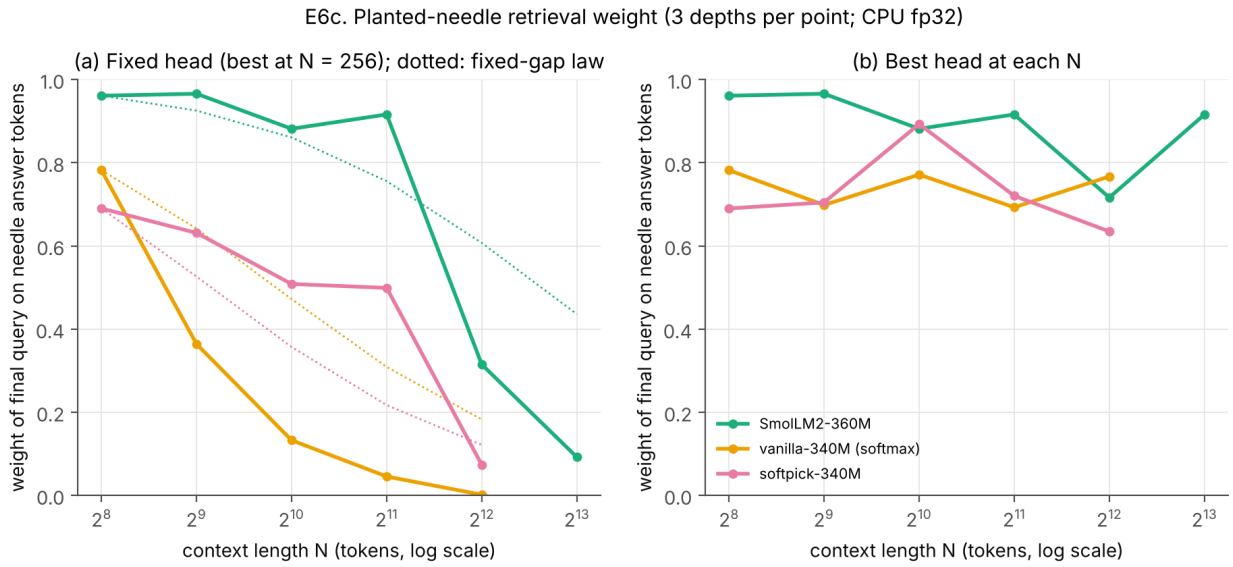


Figure 4: Needle weight for a fixed head (dotted: fixed-gap law) and for the best head at each N.

report their sink metric decreasing with sequence length (Gu et al., 2025). Softpick-340M has no sinks at any length (Table 4, Figure 3). SmolLM2-360M shows the same decline (Table 4). [**Pending:** Qwen3-0.6B and Gemma-3-1B-it: runs stopped early to save compute; rerun with `code/e6_probes.py`.]

C1a: dispersion in real heads. In vanilla-340M, the best head at 256 tokens falls from weight 0.782 to 0.0019 at 4,096 tokens, faster than the fixed-gap law. Yet the best head at each length keeps 0.767, and the model answers correctly in 15/15 cases (Figure 4). A probe that tracks one head would overstate the loss of retrieval ability.

6 Log-probability precision and serving engines

- **vLLM** returns `logits.log_softmax(dim=-1, dtype=torch.float32)` (vLLM project, 2026).
- **SGLang** casts logits to float32. Greedy requests use `log_softmax`, and standard sampling returns `torch.log(probs)` (SGLang project, 2026).
- **The exception:** with `-rl-on-policy-target` and deterministic inference, SGLang computes `log_softmax` in bfloat16.
- **The cost on synthetic logits:** with a 151,936-token vocabulary, bfloat16 `log_softmax` returns exactly 0 once the target leads by gap 17.44 ($p \gtrsim 0.996$). fp32 does so only at gap 27.88.

[**Pending:** E7: exact-zero rate on real text; B4: returned log-probs from running servers.]

7 Implications for serving

- **KV eviction.** Window-only eviction fails unless the first tokens are kept (Xiao et al., 2024). Our matched pair shows sink mass shrinking with context, and absent under softpick, so the value of pinning the first page may depend on both (inference; tested by B5).
- **Log-prob APIs.** Default paths are float32. Users of RL on-policy mode should expect exact zeros on confident tokens (established in code).
- **Training numerics.** Collapse thresholds depend on the class count, so the published constants are not safe margins for large vocabularies (established, CPU).

8 Threats to validity

- **Hardware:** CPU kernels only (B6 checks CUDA).
- **Scale:** models of 340M–1B parameters at $\leq$8,192 tokens.
- **Samples:** three passages and three depths per point, with no confidence intervals in this draft.
- **Definitions:** our sink metric averages over the last 64 queries.
- **Implementation:** the fixes are our translations of the published code.

9 Artifact availability

Code, raw JSON, logs and validation results are in `reports/softmax_paper_v3/` of the author's repository: `code/`, `data/`, `data/validate_models.json`, and GPU scripts in `code/gpu/`. B5's success criterion is pre-registered in its docstring. The survey and the census of 36 model families are in the companion report (Tata Vishnu Vardhan Rao, 2026).

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